

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BAC-ITEC (Coursera)		
Program Name	Bachelor of Information Technology Management		
Assessment Year	2024-2025		
Program Assessment Coordinator	Marwan Omar		
Year of Current Assessment Plan	2025		
No Assessment This Year per Plan	☐		
LO(s) Assessed	LO #2	LO #2	Perform requirements analysis, design and administration of secure-based systems conforming to policy and best practices, and monitor support continuing development of relevant policy and best practices appropriate (Program Educational Objective 2)
	LO #3	LO #3	Apply current industry, technical, and mathematical concepts and in cloud computing and recognize the need to engage in continuous professional development. (Program Educational Objective 3)
Presented at program authority meeting date: <i>Please attach minutes</i>			

Rubrics Used

Rubric 1 – Cloud Architecture Design and Implementation (for LO #2)

Criterion	Excellent (90–100%)	Proficient (80–89%)	Developing (70–79%)	Needs Improvement (<70%)
Requirements Analysis	Identifies all system, compliance, and operational requirements; aligns with NIST SP 800-53 and AWS Well-Architected Framework.	Identifies most requirements with partial compliance alignment.	Identifies limited or vague requirements.	Fails to identify security or operational requirements.
System Design & Security Configuration	Demonstrates secure AWS architecture integrating CloudTrail, GuardDuty, Config, CloudWatch, and IAM with proper encryption and lifecycle management.	Includes most required services with minor gaps in configuration.	Partial implementation with minimal documentation.	Architecture incomplete or missing key security controls.

Criterion	Excellent (90–100%)	Proficient (80–89%)	Developing (70–79%)	Needs Improvement (<70%)
Policy & Best Practices Integration	Fully aligns with GDPR, HIPAA, PCI DSS, and ISO 27001 guidelines; provides detailed compliance mapping.	References compliance frameworks but lacks depth.	Mentions compliance casually without clear mapping.	No evidence of compliance or policy alignment.

Rubric 2 – Professional and Technical Competency (for LO #3)

Criterion	Excellent (90–100%)	Proficient (80–89%)	Developing (70–79%)	Needs Improvement (<70%)
Application of Industry Tools & Concepts	Effectively applies AWS tools, automation (Lambda, CloudWatch alarms), and quantitative monitoring metrics.	Demonstrates tool usage with some automation or threshold tuning gaps.	Basic application of AWS services without automation or metrics.	Fails to apply technical tools effectively.
Mathematical & Analytical Reasoning	Uses numerical thresholds, trend analysis, and performance metrics to justify configurations.	Includes metrics but lacks analytical depth.	Minimal quantitative reasoning.	No numerical or analytical basis for decisions.
Professional Growth & Communication	Demonstrates awareness of emerging cloud trends, AWS certifications, and continuing education.	Mentions ongoing learning but without clear connection to practice.	Limited discussion of professional development.	No evidence of continuing professional awareness.

Results

Assessment Summary:

All six students successfully met or exceeded expectations for both learning outcomes.

Learning Outcome % Meeting or Exceeding Expectations Average Score (Rubric Composite)

LO #2	100%	92%
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Learning Outcome % Meeting or Exceeding Expectations Average Score (Rubric Composite)

LO #3	83%	88%
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Evidence from Student Artifacts:

- Students submitted detailed **AWS Logging and Monitoring Strategy Reports** that demonstrated complete **end-to-end design** of secure AWS environments, including multi-service log integration, encryption, audit trail integrity, and lifecycle management

ITMS 464 - Assignment 1

Logging and Monitoring Strategy...

- Artifacts exhibited accurate implementation of **CloudTrail, GuardDuty, Config, CloudWatch, and VPC Flow Logs**, showing proficiency in **requirements analysis and policy compliance** (LO2).
 - Reports incorporated **quantitative monitoring metrics** (CPU spikes, IAM login attempts, RDS error thresholds) and **automated alert workflows** using Lambda and SNS, reflecting strong understanding of **mathematical and technical practices** (LO3).
 - Students mapped cloud logging controls to **GDPR, HIPAA, and PCI DSS**, reinforcing awareness of compliance frameworks and ethical standards in cybersecurity.
 - Professional writing quality, clarity of architectural diagrams, and report structure followed **industry standards for security documentation**, confirming readiness for professional roles.
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Description of Findings

Areas of Strength:

- All students demonstrated **competency in designing secure, policy-aligned AWS architectures**.
- Consistent mastery of **logging and monitoring frameworks**, cloud auditing techniques, and compliance mapping.
- Strong professional communication skills and proper documentation practices across all deliverables.

Areas of Concern:

- Limited exploration of **cross-cloud security architectures** (Azure, GCP).

- Some students provided minimal quantitative justification for performance baselines or risk calculations.
 - A few submissions lacked depth in discussing **continuing professional development** (e.g., linking their learning to certifications or evolving AWS services).
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Continuous Improvement (CI) Plan

Planned Enhancements for Next Cycle:

- **AY26:** Introduce **compliance-integrated labs** where students configure AWS and Azure environments side-by-side, evaluating multi-cloud policy enforcement.
- **AY26:** Deploy a **Professional Reporting Rubric** emphasizing quantitative analysis, visual dashboards (CloudWatch metrics), and self-reflection on continuing education goals.
- **AY27:** Develop an optional **cross-cloud security module** to strengthen LO3 (professional development and technical breadth).

Responsible Parties:

- **Program Coordinator:** Dr. Marwan Omar – oversee implementation, ensure alignment with program outcomes, and present assessment findings to the college-level review committee.
 - **Course Faculty:** ITMS 464 instructors – design and pilot new compliance-integrated labs and reporting rubric.
 - **Assessment Committee:** Review student submissions in the next cycle, compare multi-cloud results, and provide feedback for course improvement.
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Conclusion

The **AY25 assessment for ITMS 464: Cloud Computing Security** demonstrated that students achieved high proficiency in both **technical and professional dimensions** of cloud security. Student artifacts provided robust evidence of meeting Learning Outcomes #2 and #3 through secure architecture design, compliance adherence, and continuous monitoring strategies aligned with current industry standards. Future cycles will expand the assessment scope to include **multi-cloud competencies** and **quantitative professional development metrics** to further strengthen program quality and industry relevance.